

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10: PROPERTIES OF WAVES (SUB-STRAND 2.1) FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Today you will write your Final Explanation of the matatu phenomenon. Think about Baraka's matatu on the Nairobi-Ngong route, and about Wanjiru and Otieno standing around the corner of a building on a busy Nairobi street. Baraka keeps his horn at 400 Hz and his radio on one 100 MHz FM station. Use the wave equation $v = f\lambda$ and the figures from our lessons: sound in air travels at 340 m/s, light in air travels at 3.0×10^8 m/s, the street opening is 4.0 m wide, and the matatu moves at a constant 15 m/s. Write your answer in the space under each prompt. For every explanation, give a claim, the evidence (with numbers and units) and your reasoning. Show your working for every calculation. Use the words pitch for frequency and loudness for amplitude. Refer back to the Driving Question Board and your class model (Parts A, B and C) where it helps.

Part A: Why the music turns the corner but the torch beam does not

Wanjiru and Otieno cannot see the matatu behind the concrete wall, but they hear its music clearly. A torch beam from the matatu would be blocked. (a) Use $v = f\lambda$ with 340 m/s to calculate the wavelength of the horn (400 Hz) and the wavelengths of the lowest (200 Hz) and highest (1000 Hz) notes of the music. (b) Calculate the ratio of the horn wavelength to the 4.0 m opening, and the approximate ratio for visible light of wavelength 5.0×10^{-7} m. (c) Write a claim-evidence-reasoning explanation of why the music is heard around the corner and the torch beam is not. (d) Which would be heard better around the corner, a 200 Hz note or a 1000 Hz note? Explain.

Write your answer here:

Part B: Why the horn changes pitch

Baraka drives past at a constant 15 m/s and never changes his horn, which has a true frequency of 400 Hz. (a) Use $f' = f v / (v - v_s)$ and $f' = f v / (v + v_s)$ with $v = 340$ m/s to calculate the frequency Otieno hears as the matatu approaches and the frequency Wanjiru hears as it moves away. Round to the nearest Hz. (b) Find the drop in pitch. (c) Explain, using wavefronts, why the pitch heard changes although the horn stays the same. (d) Which fixed quantity stays the same at the horn itself?

Write your answer here:

Part C: Why the FM radio fades near the Ngong Hills

The matatu radio is tuned to one 100 MHz FM station that never changes, yet it fades and crackles at certain spots near the Ngong Hills and comes back a few metres later. (a) Use $v = f\lambda$ with 3.0×10^8 m/s to find the wavelength of the signal. (b) Explain, using the words superposition, path difference and interference, why the signal fades at some spots and is strong between them. (c) How far apart are the weak spots, and how does this follow from your answer to (a)? (d) In the classroom, two speakers both played 340 Hz. Give the wavelength and say whether a path difference of 1.00 m or 0.50 m gave the quiet spot. How does this support your explanation in (b)?

Write your answer here:

Part D: Using wave ideas in technology

Wave properties are applied in everyday life. (a) Fishers on Lake Victoria use sonar. A pulse returns after 0.040 s and sound travels at 1500 m/s in water. Calculate the depth and explain why you divide by 2. (b) A doctor uses ultrasound of frequency 3.5 MHz, which travels at 1540 m/s in soft tissue. Calculate its wavelength in millimetres and state how the image is formed. (c) In FM radio, a 1 kHz tone is sent on a 100 MHz carrier. State what changes in the carrier and what stays constant.

Write your answer here:

Part E: Putting it all together

Write a complete explanation for the Driving Question: How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves? Your answer must have three paragraphs matching Parts A, B and C of the class model. In each paragraph give a claim, evidence with numbers from this assessment, and reasoning. Then say how your final model differs from your first model in Lesson 1.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining diffraction at the corner (Part A)	Correctly calculates all wavelengths and ratios (0.85 m, 1.7 m, 0.34 m, 0.21, about 1.3×10^{-7}) and gives a full claim-evidence-reasoning explanation linking wavelength to gap size, including why low pitches bend more than high pitches.	Calculates most wavelengths and ratios correctly and explains that sound bends and light does not because of wavelength compared with the opening, with small gaps in reasoning.	States that sound bends and light does not but gives little or no use of wavelength compared with the 4.0 m gap, or has calculation errors.
Explaining the Doppler effect (Part B)	Uses both Doppler formulas correctly to get 418 Hz, 383 Hz and a 35 Hz drop, and clearly explains compressed and stretched wavefronts while the horn stays at 400 Hz.	Gets the correct frequencies and gives a mostly correct explanation of higher and lower pitch with some vagueness about wavefronts.	Gives incorrect values or says the horn itself changes, or does not link pitch change to motion.
Explaining interference and fading (Part C)	Finds $\lambda = 3.0$ m, uses path difference ($n\lambda$ and $(n+1/2)\lambda$) correctly, links to the 1.5 m spacing, and connects it to the 340 Hz speaker results.	Finds the wavelength and explains destructive and constructive interference of direct and reflected signals, with minor errors in path difference or spacing.	Mentions weak signal without explaining the overlap of direct and reflected waves, or gives an incorrect wavelength.
Calculations and application to technology (Part D)	Correctly calculates 30 m, 0.44 mm and describes FM modulation, with units and explanations of $d = vt/2$.	Gets most calculations correct with units and a basic description of reflection and modulation.	Calculation errors, missing units, or does not explain the division by 2.
Scientific communication and synthesis	Uses claim, evidence and reasoning in every part, uses correct terms and units, uses Kenyan context, and compares the final model with the Lesson 1 model.	Mostly uses claim, evidence and reasoning with correct terms and units, with some weak links between parts.	Answers are incomplete or lack evidence and reasoning, with frequent errors in terms or units.

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